Luis Victoria

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Work Experience

Software Engineer — Market Data Platform, Capacity & Performance Feb 2023 – Jul 2026
Bloomberg LP *New York, NY*

Engineered performance, capacity, and reliability systems for Bloomberg's latency-sensitive, distributed Market Data Platform, spanning Feeds and the Ticker Plant. The platform handled >700B ticks/day during extreme volatility while maintaining <6 ms median and <250 ms p99.9 latency SLAs.

- **Capacity Planning & Cluster Rollouts**: Assessed CPU, memory, and storage headroom ahead of onboarding new global exchanges; owned staged multi-region rollouts of cluster splits and new clusters, rebalancing workloads off overloaded machines to avoid SLO breaches with minimal client disruption.
- **Workload Prioritization & Failover**: Automated load reassignment across data centers during failover; benchmarked CPU thread-pinning under combined load, informing Linux cgroups-based CPU prioritization that preserved critical broadcast traffic while bounding dropped requests.
- **Predictive Capacity Forecasting**: Rewrote a fragile legacy log-scraping storage monitor as a two-part Python system: a fleet-wide scanner writing structured per-database storage and machine metadata to a centralized datastore, and a forecasting service extrapolating rolling usage trends 14 days ahead with automated alerting. Delivered in one week during the April 2025 tariff-driven volatility spike, protecting 1,000+ machines from storage-exhaustion/data-loss risk.
- **Performance Regression Detection**: Built a statistical regression-detection pipeline on Kubernetes-native Argo Workflows, applying Bayesian Online Changepoint Detection to flag anomalous processing latencies across 30,000+ tick processor instances; cut remediation time from days to <24 hours.
- **Latency Instrumentation & Observability**: Authored a C++20 envelope-based timestamping framework tracing tick updates stage-by-stage through the Ticker Plant pipeline with µs resolution, enabling latency regression detection and optimization with <10 µs overhead.
- **Production & Incident Ownership**: On-call engineer in 24/7 follow-the-sun rotation for Bloomberg's Market Data Platform; isolated bad data, performed live synchronization and recovery, and drove root-cause analysis for major outages under volatile market conditions.

Quantitative Developer Intern May 2019 – Aug 2019
Farringdon Asset Management *Singapore*

- **Automation with Bloomberg API**: Reduced time to generate client portfolio analysis by 50% for the portfolio management team using information from PORT<GO>.
- **Backtesting Enhancements**: Extended the firm's Python backtesting engine to support cryptocurrency analysis, integrating market-data pipelines through CCXT and Binance.

Education

Business and Computer Science Sep 2018 – Dec 2022
Bachelor of Commerce, Sauder School of Business, University of British Columbia *Vancouver, BC*

- **Relevant Coursework**: Machine Learning & Data Mining, Matrix Algebra, Calculus III, Empirical Economics, Advanced Algorithmic Design, Advanced Relational Databases.
- **Sauder Trading Simulation**: Built a Raspberry Pi arbitrage bot for a non-matching order-book simulation, polling every 2 min and executing crossed bid/ask opportunities; placed 1st of 136 students.

Skills

Programming Languages: C++20, Python, SQL (PostgreSQL)
Tools: Linux, Grafana, Humio (LogScale), Jenkins, Git, pytest, pydantic, uv, Ruff, Argo Workflows